MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys). My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.

This work aims to study, from an academical perspective at first, the subject of continuous measurement considering open quantum systems which was a different topic from my supervisor's specific area of research. Our primary motivation was therefore initially to educate ourselves on a field of quantum physics that is well known to a large part of the scientific community. By familiarizing ourselves with the topic, we decided to adopt a particular frame of report due to our journey of understanding. Therefore, the structure of this document is meant to help discovering and exploring the fundamentals of quantum continuous measurement to any unfamiliar reader. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts in the chapter concerning continuously monitored quantum open systems.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in space A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

I From closed systems to continuous measurements — in a nutshell

In this chapter we will briefly summarize the definitions needed for continuously monitored open quantum systems formalism.

I.i) Generalities on quantum mechanics

I.i.1⟩ States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H}. \tag{1}$$

The Hamiltonian $\hat{H}$ is a specific Hermitian[2] operator representing classically the total energy of a system; this relationship will be detailed in Section I.i.2⟩. Physically, the evolution of a system is governed by its energy, the *Schrödinger equation* states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle\,. \tag{2}$$

We can define a unitary[3] operator $\hat{U}$ such that,

$$|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle\,, \tag{3}$$

solving formally the Schrödinger equation. By injecting the previous relation in the state equation, we get

$$\frac{\mathrm{d}\hat{U}}{\mathrm{d}t}(t,t_0) = -\frac{i}{\hbar}\hat{H}(t)\hat{U}(t,t_0). \tag{4}$$

In the general case, the Hamiltonian can depend on time, so we need the chronological time-ordering exponential [3] to solve Eq. (4). However, in the simplest case, the Hamiltonian is independent of time and the solution becomes

$$|\psi(t)\rangle = \underbrace{e^{-\frac{i\hat{H}(t-t_0)}{\hbar}}}_{\hat{U}(t,t_0)}\,|\psi(t_0)\rangle\,. \tag{5}$$

This solution of the Schrödinger equation is indeed **unitary** and we can use $\hat{U}^\dagger(t,t_0)$ on the left of both sides of Eq. (5) to obtain $|\psi(t_0)\rangle = \hat{U}^\dagger(t,t_0)\,|\psi(t)\rangle$, which proves that the dynamics of closed systems are **reversible**.

I.i.2⟩ Observables

In quantum mechanics, any physical quantity is represented by an **observable** which is an operator fulfilling two properties: the eigenvectors of the observable form an orthonormal basis of the Hilbert space $\mathcal{H}$, and, the corresponding eigenvalues are necessarily real [2].

[2] $\hat{H}^\dagger = \hat{H}$, where † denotes the conjugate transpose, defining the Hermitian adjoint.

[3] $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{I}$, where $\mathbb{I}$ is the identity.

In finite-dimensional Hilbert spaces, the definition of Hermitian operator[4] is sufficient for it to be an observable. Moreover, in the case of non-finite dimension Hilbert spaces, the definition of an observable requires more rigorous mathematical foundations (notably self-adjointness property), the details of which lie beyond the purview of this study. In this sense and to lighten notations, we will only consider finite dimension cases in the following.

Let us denote $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document; furthermore, $\hat{\bullet}$ symbols on operators are omitted hereafter for notational convenience. A natural operator whose eigenbasis $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ is particularly significant is the Hamiltonian H, as its eigenvalues $\{E_n\}_{n\in I_{\mathcal{H}}}$ correspond to the allowed energy values of a system. So, the Hamiltonian is well-described by its corresponding eigenequation

$$\forall n \in I_{\mathcal{H}}, \quad H\,|\phi_n\rangle = E_n\,|\phi_n\rangle\,. \tag{6}$$

Since $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ forms a basis of the Hilbert space, one can describe any state $|\psi\rangle$ in this very basis:

$$\forall\,|\psi\rangle \in \mathcal{H}, \qquad |\psi\rangle = \sum_{n\in I_{\mathcal{H}}} \overbrace{c_n}^{\langle\phi_n|\psi\rangle}\,|\phi_n\rangle\,. \tag{7}$$

This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian.

Finally, let us mention the normalization of a state which is important to handle scalar products properly. To normalize a state, we set $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$, the condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$ being thus equivalent.

I.i.3⟩ Purity of states

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I.i.4⟩ Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathrm{End}(\mathcal{H})$ respecting Properties **??**. For pure states we can describe them $|\psi\rangle \in \mathcal{H}$, or, by using the pure matrix density formalism $\rho_p = |\psi\rangle\langle\psi|$.

2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.

[4] The eigenvectors of a Hermitian operator always span the entire Hilbert space (Spectral Theorem). Using the Gram-Schmidt process, we can always find a complete orthonormal basis from the operator's eigenspaces.

4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}[a_n]$ (which depends on the spectrum of the observable).

5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement collapses into the eigenstate $|\phi_n\rangle$.

6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the *Schrödinger* equation:
$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = -\frac{i}{\hbar} H(t)\,|\psi(t)\rangle \tag{8}$$
For density matrices, the evolution is driven by the *Liouville-von Neumann* equation[5]:
$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\,[H(t), \rho(t)] \equiv \mathcal{L}(t)\rho(t) \tag{9}$$

[5] $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is the commutator of $\hat{A}$ and $\hat{B}$

I.ii) Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution, however, this description applies to the ideal case where the quantum system is considered isolated from any external interaction. That is why we will introduce in this section the concepts of OQS.

In the OQS theory, a system is considered to interact locally with its environment, which itself interacts with a wider environment, etc... To avoid considering infinite chains of interaction, the theory truncates the environment to a local vicinity of the system. Thus, we approximate the total system T as a closed quantum system, whereas the quantum system of interest S couples with its environment E and is called a quantum open system. This coupling with the environment thus gives rise to a dynamical regime distinct from that of closed quantum systems. Fundamentally, the Hilbert space of study is divided into $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$.

REWRITE

♦ The *Born* approximation consists in saying that the environment acts as a thermodynamical reservoir, *i.e.*, the environment is so big compared to the system that the interaction does not alter its state:

$$\tilde{\rho}_T(t) = \tilde{\rho}_S(t) \otimes \rho_E \tag{10}$$

It describes a so-called "weak-coupling" between the system and the environment.

♦ The *Markov* approximation assures that the entire system has no memory. That is to say, the environment is so rapidly agitating that it dissipates information of the system instantaneously. This implies that the environment comes back to equilibrium instantaneously after each interaction. Physically, the timescale of the reservoir correlation function decay τ_E is smaller than the timescale τ_R on which the system varies "appreciably".

These two last approximations lead to the *Markovian quantum master equation.*

♦ The *secular* approximation allows the possibility to neglect rapidly oscillating terms in the quantum *Markov* master equation.

I.iii) Quantum Measurement Theory (QMT)

In the QMT [6, 7], measuring consists in applying a projector to a state, so, we call this process a *projective measurement*. The set of projective measurements is called a Projection-Valued Measure (PVM) [8]. For the sake of simplicity, we work — as until now — with non-degenerate spectra for the observables. Since an observable has a set of eigenvectors $\{|a_n\rangle\}_{n\in I_{\mathcal{H}}}$, we can build a projector corresponding to one of them:

$$P_{a_n} = |a_n\rangle\langle a_n| \,. \tag{11}$$

This definition implies four important properties:

$$P_{a_n}^\dagger = P_{a_n}, \qquad P_{a_n}^2 = P_{a_n}, \qquad P_{a_n}P_{a_m} = P_{a_n}\delta_{nm}, \qquad \sum_n P_{a_n} = \mathbb{I}.$$

To represent the quantum mechanics definition of a measurement, the projective measurement needs to fulfill some requirements. The first requirement is to generate the probability of observing the result of a measurement on a system. Thus, the probability to observe a_n is defined by

$$\mathbb{P}_{|\psi\rangle}[a_n] = \langle\psi|P_{a_n}|\psi\rangle = |\langle\psi|a_n\rangle|^2, \tag{12}$$

$$\mathbb{P}_{\rho}[a_n] = \mathrm{Tr}\left[\rho P_{a_n}\right], \tag{13}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

$$|\tilde{\psi_a}\rangle = P_a\,|\psi\rangle\,, \tag{14}$$

$$\tilde{\rho_a} = P_a\rho P_a. \tag{15}$$

Here the $\tilde{\bullet}$ symbol means that the state is not normalized, which will be an important notation for later, see Section **??**. By normalizing we find the proper expression of a state after a measurement with result a_n:

$$|\psi_a\rangle = \frac{P_a\,|\psi\rangle}{\sqrt{\langle\psi|P_a|\psi\rangle}}, \tag{16}$$

$$\rho_a = \frac{P_a\rho P_a}{\mathrm{Tr}\left[\rho P_a\right]}. \tag{17}$$

The obtained state is said **conditioned by the measurement** result a_n, hence the importance of the index a_n on states obtained after measurement. We point out that the normalization factor is exactly the probability to observe the result a_n for the density matrix (or the square root of the probability, when using ket formalism). In the end, projective measurement fulfill the properties of quantum mechanics by construction because they generate probabilities (respecting the fact they need to be positive and their sum is 1), and can also be used to write the expression of collapsed states after measurement.

However, the definition of PVM has been found too restrictive to represent every kind of measurement allowed by quantum mechanics [9, 10]. One can define a more general kind of measurement, that fulfills quantum mechanics needs while relaxing some properties. In particular, the orthogonality of measure elements is not required in order to generate

probabilities. Therefore, one can build the so-called Positive Operator-Valued Measure (POVM), implicitly defined by the probability to observe the result a_n:

$$\mathbb{P}_\psi[a_n] = \langle\psi|\Pi_{a_n}|\psi\rangle\,, \tag{18}$$
$$\mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho\Pi_{a_n}\right], \tag{19}$$

where the operator Π_{a_n} is called the *effect* [7] for the result a. Moreover, the set of all effects $\{\Pi_{a_n}\}_{n\in I_\mathcal{H}}$ forms the POVM. As stated, the only properties we force the POVM to respect are:

$$\text{positive semi-definite: } \forall\,|\phi\rangle \in \mathcal{H},\ \langle\phi|\Pi_{a_n}|\phi\rangle \geq 0,\ \text{and,} \qquad \sum_{n\in I_\mathcal{H}} \Pi_{a_n} = \mathbb{I}.$$

These properties imply that POVM generate proper probabilities. Furthermore, the (normalized) states obtained after finding the result a_n are now,

$$|\psi_{a_n}\rangle = \frac{K_{a_n}\,|\psi\rangle}{\sqrt{\langle\psi|\Pi_{a_n}|\psi\rangle}}, \tag{20}$$
$$\rho_{a_n} = \frac{K_{a_n}\rho K_{a_n}^\dagger}{\mathrm{Tr}\left[\rho\Pi_{a_n}\right]}, \tag{21}$$

where we introduced the operator K_{a_n} called a *Kraus operator*; they respect the relation $\Pi_{a_n} = K_{a_n}^\dagger K_{a_n}$.

Also, POVM, just like PVM, allow one to describe the process of quantum measurement while *ignoring the results*. As in the thought experiment **??** for statistical mixture of states, we can describe statistically the state without knowing the measurement results:

$$\rho = \sum_{n\in I_\mathcal{H}} \mathbb{P}[a_n]\rho_{a_n} = \sum_{n\in I_\mathcal{H}} K_{a_n}\rho K_{a_n}^\dagger. \tag{22}$$

The corresponding relation for PVM is obtained by replacing both K_{a_n} and $K_{a_n}^\dagger$ by P_{a_n}.

Finally, one can observe that POVM generalize PVM in the sense that projectors are special cases of POVM elements (adding self-adjoint property and orthogonality to POVM gives a PVM). Furthermore, PVMs make states collapse on eigenstates that are necessarily orthogonal to each other, which is not necessarily the case for POVMs. A brief example of POVM forcing a state to collapse onto "non-orthogonal states" is when considering initially the state $|\psi\rangle = \frac{|g\rangle+|e\rangle}{\sqrt{2}} \equiv |+\rangle$. If we choose the Kraus operators $K_0 = |g\rangle\langle g|$ and $K_1 = |+\rangle\langle e|$, which fulfill $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$, and, that form the POVM $\{|g\rangle\langle g|\,, |e\rangle\langle e|\}$. Then, when the result of the measurement is 0, we find that the state collapses to $|\psi_0\rangle = |g\rangle$, and when finding 1 it collapses to $|\psi_1\rangle = |+\rangle$. However, $\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}} \neq 0$, thus the states on which the initial state can collapse are not orthogonal to each other.

II Continuously measured quantum systems – obtaining a Stochastic Master Equation (SME)

A Representations of quantum mechanics

Brief complement of quantum mechanics

An average is defined in quantum mechanics by

$$\begin{aligned}\langle O\rangle_{\psi} &= \langle\psi|O|\psi\rangle\,,\\ \langle O\rangle_{\rho} &= \mathrm{Tr}\left[\rho O\right],\end{aligned}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

A.a) Schrödinger and Heisenberg pictures

In this appendix we denote the pictures with their acronyms and lighten the notation of the total state $\rho_T \equiv \rho$. However, this could be misleading because S in this appendix stands for *"in the Schrödinger picture"*, whereas, S in the main chapters means *"state of the system S"*. Nevertheless, we only consider the state in a closed quantum system in the following so we cannot encounter that kind of situation.

In the Schrödinger picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$, where $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. Nevertheless, the following results can be derived in the general case of a time dependent Hamiltonian, [3]. Considering the average value of an observable $A_S(t)$ in the Schrödinger picture, we find

$$\begin{aligned}\langle A_S(t)\rangle &= \langle\psi(t)|A_S|\psi(t)\rangle\\ &= \langle\psi|\underbrace{U^{\dagger}(t)A_S(t)U(t)}_{A_H(t)}|\psi\rangle\,.\end{aligned} \tag{A. 1}$$

This calculation naturally reveals the Heisenberg representation of observables and states. Indeed, we have that $\forall t,\ |\psi(t)\rangle_H = |\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time. Furthermore, we can link both representation by calculating the evolution of an observable in the Heisenberg picture:

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}}. \tag{A. 2}$$

This link can even lead to the famous *Ehrenfest theorem* for quantum mechanics:

$$\boxed{\frac{\mathrm{d}\langle A\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\langle[H(t), A(t)]\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle}. \tag{A. 3}$$

We observe that if the observable doesn't depend explicitly on time (thus constant), then the dynamics of the operator are fully described by the Heisenberg Eq. (A. 2).

A.b) Interaction picture

As mentioned in I.ii), we use perturbation theory to describe the interaction between the system S and the environment E,

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I, \tag{??}$$
$$U(t, t_0) = U_0(t, t_0) U_I(t, t_0). \tag{??}$$

Because $|\psi(t)\rangle_S = U(t)\,|\psi\rangle_S$, we can write that

$$\rho_S(t) = U(t)\rho_S U^\dagger(t). \tag{A. 4}$$

If we now try to calculate the average of an operator (in the total space $\mathcal{H}_T$) using the Schrödinger picture, we can find a new interesting relation by injecting Eq. (A. 4,**??**) in the average:

$$\begin{aligned}\langle O_S(t)\rangle &= \mathrm{Tr}\,[\rho_S(t) O_S(t)] \\ &= \mathrm{Tr}\,[\underbrace{U_I(t)\rho_S U_I^\dagger(t)}_{\tilde{\rho}(t)}\ \underbrace{U_0^\dagger(t) O_S(t) U_0(t)}_{\tilde{O}(t)}].\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\rho}(t) = U_I(t)\rho_S U_I^\dagger(t), \tag{??}$$
$$\tilde{O}(t) = U_0^\dagger(t) O_S(t) U_0(t). \tag{??}$$

We can now reverse Eq. (A. 4) and then use Eq. (**??**):

$$\rho_S = U^\dagger(t)\rho_S(t)U(t) = U_I^\dagger(t)U_0^\dagger(t)\rho_S(t)U_0(t)U_I(t),$$

and, by injecting this into Eq. (**??**), we get

$$\begin{aligned}\tilde{\rho}(t) &= U_I(t)U_I^\dagger(t)U_0^\dagger(t)\rho_S(t)U_0(t)U_I(t)U_I^\dagger(t) \\ &= U_0^\dagger(t)\rho_S(t)U_0(t).\end{aligned} \tag{??}$$

So that we find the results used in the main part of this work

$$\boxed{\tilde{\rho}(t) = U_I(t)\rho_S U_I^\dagger(t) = U_0^\dagger(t)\rho_S(t)U_0(t)}. \tag{A. 5}$$

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